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ECH512 – ANALYTICAL TECHNIQUES

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. The radiation having the shortest wavelength is
   a) Microwave b) Infrared c) Ultraviolet d) Visible

2. In IR spectrometer, the cell container is made up of
   a) Glass b) Quartz c) Alkali halide d) None of these

3. Raman effect is due to _____.
   a) Absorption of light b) Emission of light
   c) Inelastic scattering of light d) Both (a) and (b)

4. NMR spectrum is observed in _____ region.
   a) Radiofrequency b) Microwave
   c) Ultraviolent d) X ray

5. The correct order for the basic features of a mass spectrometer is
   a) Acceleration, deflection, detection, ionization
   b) Ionization, acceleration, deflection, detection
   c) Acceleration, ionization, deflection, detection
   d) Acceleration, deflection, ionization, detection

6. Mossbauer spectroscopy is the study of emission and absorption of _____rays.
   a) Alpha b) Gamma c) Beta d) UV

7. Which of the following is used for measuring the intensity of transmitted radiation?
   a) Monochromator b) Amplifier
   c) Detector d) Filter

8. The most common source used in Atomic Absorption Spectroscopy is_____
   a) Mercury lamp b) Deuterium lamp
   c) Hallow cathode lamp d) Quartz

9. _____ electrode is used in pH measurements.
   a) Calomel b) Platinum
   c) Quinhydrone d) None of these

10. Instrument used for the measurement of fluorescence is known as _____.
    a) Spectrofluorimeters b) Spectrophotometry
    c) Flame photometry d) Colorimetry

II. Fill up the blanks

11. _____is not used as a solvent in IR spectroscopy.

12. The ultraviolet region is _____nm.

13. ESR spectroscopy is also known as _____.

14. _____is used as a detector in flame photometer.

15. _____is defined as the fraction of the incident radiation which is re-emitted as fluorescence.

III. Match the following

16. Radiation Source - Read out device

17. Detector - Photomultiplier
18. Monochromator - Deuterium Lamps

19. Recorder - Filter

20. Finger print region - Mass spectroscopy

- IR spectroscopy

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Why is methanol used as a good solvent for UV but not for IR spectroscopy?

22. State the Beer Lambert's law.

23. Calculate the frequency of a radiation whose wavelength is 420 nm.

24. Write the advantages of Raman spectroscopy over IR spectroscopy.

25. Why TMS is used as a reference standard in NMR spectroscopy?

26. Name the radiation source and detector used in NMR and ESR.

27. Explain the principle of ESR spectroscopy.

28. Define atomization.

29. What is flame photometry? Name few metals which can be detected by this method.

30. What is filter? Name two types of filters.

31. Compare Atomic Absorption Spectroscopy and Flame photometry.

32. Give any two applications of Fluorimetry.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) What are monochromators? Name the principle parts of it.
    b) Draw a schematic diagram of an ultraviolet spectrophotometer and explain the specific functions of various components.

34. a) Discuss briefly about a single-beam and double beam spectrometer in IR spectroscopy.
    b) Write the advantages for the single beam spectrometer.

35. Write short note on the following
    a) Sample cavity and detector in NQR spectroscopy.
    b) Crystal detector in ESR spectroscopy.

36. Explain the principle and instrumentation of atomic absorption spectroscopy with the neat diagram.

37. Outline the instrumentation of nephlometry and turbidimetry.

38. Describe briefly the optical system and detectors used in flame photometry.

39. a) Write any one method to determine the pH of a solution.
    b) Explain the instrumentation of fluorimetry.

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